1 :- What is need for Hypothesis Testing ?

Ans :- The need for Hypothesis Testing arises when we want to make decisions or draw conclusions about a population based on sample data. In statistics, it provides a structured way to test assumptions using evidence rather than guesswork.

Here are the main reasons why hypothesis testing is important:

1. Decision Making

Hypothesis testing helps in making logical decisions based on data. Example: A company can decide whether a new product is better than the old one. 2. Testing Assumptions

It allows us to check whether an assumption (hypothesis) about a population is true or not.

Null hypothesis (H_0): No effect or no difference Alternative hypothesis (H_1): There is an effect or difference

3. Scientific Research

In fields like Statistics and Data Science, hypothesis testing is essential for validating theories and experiments. 4. Reducing Uncertainty

Instead of relying on intuition, hypothesis testing uses probability to reduce uncertainty and provide more reliable conclusions. 5. Quality Control

Industries use it to check product quality and maintain standards. Example: Testing whether a batch of products meets required specifications. 6. Comparing Groups

It helps compare two or more groups to see if differences are significant. Example: Comparing student performance under two teaching methods.

7. Supports Data-Driven Decisions

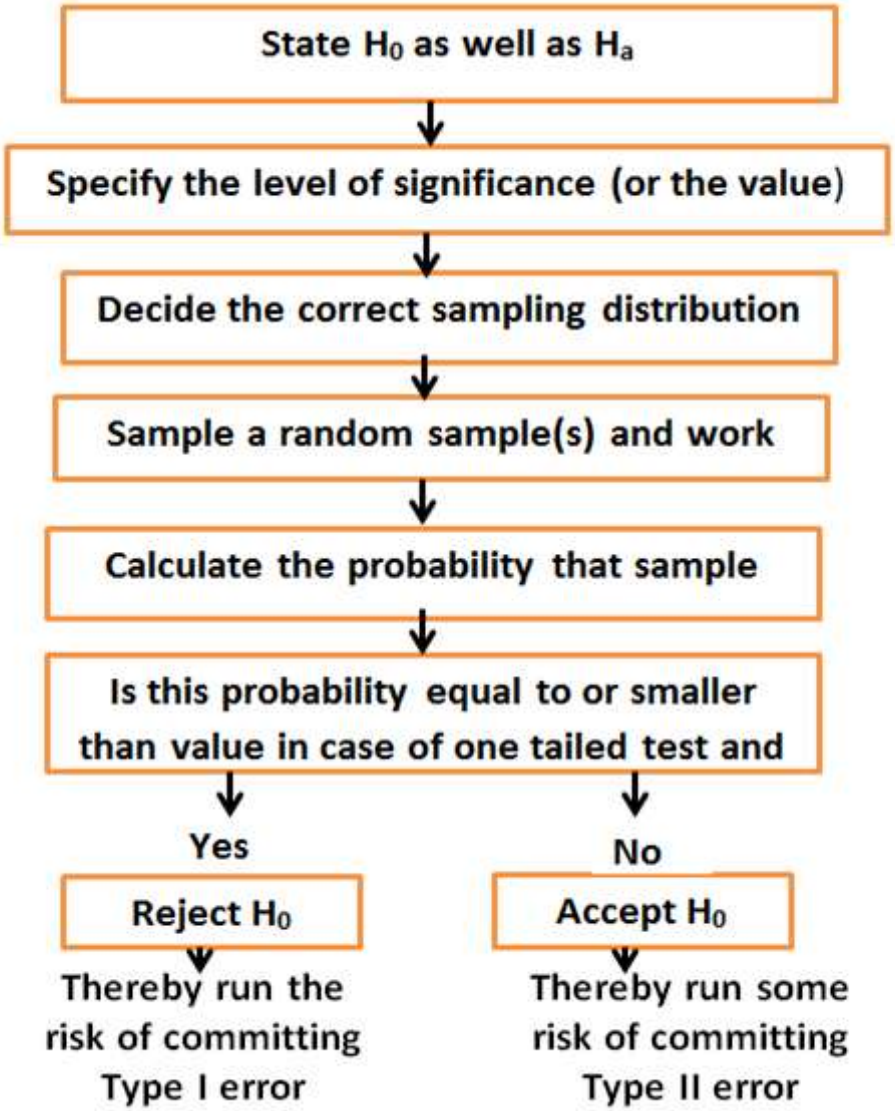
It forms the basis of modern analytics and business intelligence, ensuring decisions are backed by data.

2 :- With a neat diagram show important steps in hypothesis Testing.

Ans :- 1. Formulate Hypotheses Null Hypothesis (H_0): No effect / no difference Alternative Hypothesis (H_1): There is an effect / difference

2. Select Significance Level (α) Common values: 0.05 (5%), 0.01 (1%) It represents the risk of rejecting a true H_0

3. Choose Appropriate Test Z-test, t-test, Chi-square test, etc. Depends on data type and sample size
4. Collect Data & Calculate Test Statistic Use sample data to compute value (Z, t, etc.)
5. Determine Critical Value / p-value Compare calculated value with critical value OR Compare p-value with α
6. Make Decision If test statistic > critical value → Reject H_0 If p-value < α → Reject H_0 Otherwise → Fail to reject H_0



Conclusion Interpret the result in simple words for decision-making

3 :- What is a student t distribution?

Ans :- Student's t-distribution is a continuous probability distribution used in statistical analysis when the sample size is small and the population standard deviation is unknown.

It was developed by William Sealy Gosset.

Formula

$t = s/n\bar{x} - \mu$

Characteristics:

It is bell-shaped and symmetrical.

It has heavier tails than the normal distribution.

It depends on degrees of freedom (df = n – 1).

As sample size increases, it approaches the normal distribution.

Uses:

Used for small sample sizes (n < 30)

Applied when population standard deviation is unknown

Used in hypothesis testing (t-test)

Helps in confidence interval estimation

Conclusion:

Student's t-distribution is an important statistical tool used to make reliable inferences about a population when only a small sample is available.

4 :- What is level of significance ? What role t plays in hypothesis Testing

Ans :- The level of significance (α) is the probability of rejecting the null hypothesis (H_0) when it is actually true.

In simple terms, it is the maximum acceptable error in hypothesis testing. Common Values:

a = 0.05(5%)

a = 0.01(1%)

Importance:

It helps in decision-making during hypothesis testing.

It defines the critical region (rejection region).

It controls the Type I error (false rejection of H_0).

Role of t in Hypothesis Testing

The t-value comes from the Student's t-distribution, developed by William Sealy Gosset.

The t-statistic is used to test hypotheses about the population mean when:

Sample size is small Population standard deviation is unknown

Formula:

t = s / n x ¯ − μ

Role in Hypothesis Testing:

Calculates Difference

Measures how far the sample mean is from the population mean.

Decision Making

Compare calculated t value with critical t value

OR compare p-value with α

Used in t-tests

One-sample t-test

Two-sample t-test

Paired t-test

Works with Small Samples

Provides reliable results when data is limited.

Conclusion:

Level of significance (α) sets the rule for accepting or rejecting H_0.

The t-statistic helps in making that decision by comparing calculated values with critical limits.

5 :- Design a hypothesis testing for identifying average age of students in TE IT class is 19 or not.

Ans :- Problem: To test whether the average age of students in TE IT class is 19 or not.

Step 1: Formulate Hypotheses Null Hypothesis (H_0):

μ =19 μ=19 (Average age is 19) Alternative Hypothesis (H_1):

μ ≠19 μ =19 (Average age is not 19) Step 2: Level of Significance Choose

α = 0.05

α=0.05 (5% level of significance)

Step 3: Select Test

Since population standard deviation is unknown and sample size may be

small → use Student's t-test

(Developed by William Sealy Gosset) Step 4: Test Statistic

t

$\bar{x}$ − 19 s / n t= s/ n

x ¯ −19

Where:

$\bar{x}$ x ¯ = sample mean age s s = sample standard deviation n n = number of students Step 5: Decision Rule Find critical t-value from t-table with

$d\ f$

n − 1 df=n−1 at α

0.05 α=0.05 (two-tailed) Decision: If | t |

$t\ c\ r\ i\ t\ i\ c\ a\ l$ |t|>t critical

→ Reject H_0 If | t | ≤ $t\ c\ r\ i\ t\ i\ c\ a\ l$ |t|≤t critical

→ Fail to reject H_0

```
import numpy as np
from scipy import stats

# Sample data: ages of TE IT students
ages = [18, 19, 20, 19, 18, 21, 19, 20, 18, 19]

# Given population mean
mu = 19

# Perform one-sample t-test
t_statistic, p_value = stats.ttest_1samp(ages, mu)

print("t-statistic:", t_statistic)
print("p-value:", p_value)

# Level of significance
alpha = 0.05
```

```
# Decision
if p_value < alpha:
    print("Reject H0: Average age is not 19")
else:
    print("Fail to Reject H0: Average age is 19")
```

```
t-statistic: 0.3179993640019125
p-value: 0.757740072772952
Fail to Reject H0: Average age is 19
```

6 :- Check if you can accept Ho at alpha 95% ? Match your conclusions with python functions.

Ans :- Problem: Check whether we can accept H_0 (average age = 19) at 95% level of significance and verify using Python.

Step 1: Hypotheses H 0 : μ

19 H 0

:μ=19 H 1 : $\mu \neq$ 19 H 1

:μ =19 (two-tailed test) Step 2: Level of Significance 95% confidence ⇒ α= 0.05 α=0.05 Step 3: Sample Data (Example)

Ages of students: 18, 19, 20, 19, 18, 21, 19, 20, 18, 19 Step 4: Test Statistic After calculation (approx): x ¯

19.1 x ¯ =19.1 t ≈ 0.33 t≈0.33 d f

9 df=9 Step 5: Critical Value From t-table at:

α

0.05 α=0.05, two-tailed d f

9 df=9 Step 6: Decision Calculated t

0.33 t=0.33 Since | 0.33 | < 2.262 |0.33| Step 7: Conclusion At 95% confidence level, there is no significant difference.

```python
import numpy as np
from scipy import stats

ages = [18, 19, 20, 19, 18, 21, 19, 20, 18, 19]
mu = 19

t_stat, p_value = stats.ttest_1samp(ages, mu)

print("t-statistic:", t_stat)
print("p-value:", p_value)

alpha = 0.05

if p_value < alpha:
    print("Reject H0")
else:
    print("Fail to Reject H0")
```

```
t-statistic: 0.3179993640019125
p-value: 0.757740072772952
Fail to Reject H0
```